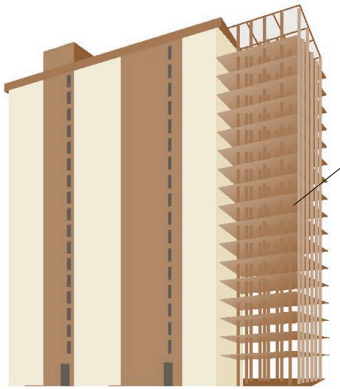
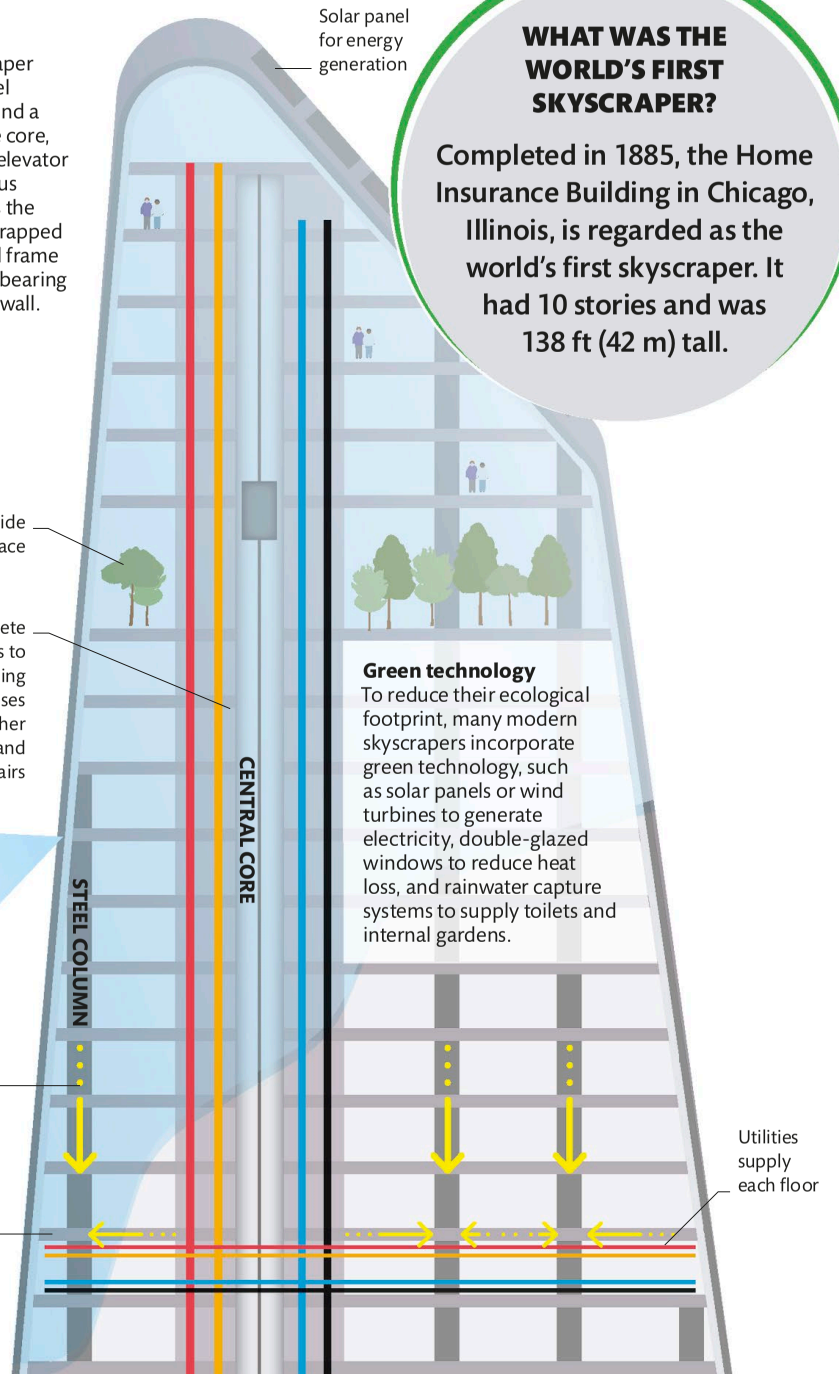
WOODEN SKYSCRAPERS

New types of engineered wood have made it possible to construct high-rise buildings with wooden frames. Made from thin layers of wood crisscrossed and stuck together with glue, this engineered wood—known as glulam—is as strong as steel. Existing wooden-framed skyscrapers include the 18-story Brock Commons student residence in Vancouver.



BROCK COMMONS, VANCOUVER

Anatomy of a skyscraper
A typical skyscraper consists of a steel framework around a central concrete core, which contains elevator shafts and various services, such as the water supply. Wrapped around the steel frame is the non-load-bearing exterior curtain wall.

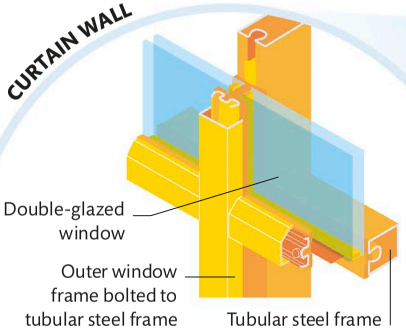


Green technology
To reduce their ecological footprint, many modern skyscrapers incorporate green technology, such as solar panels or wind turbines to generate electricity, double-glazed windows to reduce heat loss, and rainwater capture systems to supply toilets and internal gardens.

WHAT WAS THE WORLD'S FIRST SKYSCRAPER?

Completed in 1885, the Home Insurance Building in Chicago, Illinois, is regarded as the world's first skyscraper. It had 10 stories and was 138 ft (42 m) tall.

CURTAIN WALL



The exterior walls—called curtain walls—are attached to the building's framework. The walls do not bear the weight of the building and need to support only their own weight.